



Programming Fundamentals

G-LAB # 1

Instructions:

- You have to submit CPP files of the program in a folder, named your Registration Number.
 - You have to upload your g-lab task on CMS.
 - All program codes should be written in C/C++. Students should use Visual studio compiler for coding.
 - Indent and comment your code.
 - Use meaningful variable names
 - Plan your code carefully on a piece of paper before you implement it.
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Question :

Task Description:

You need to develop a **Vowel Frequency Analyzer** that processes text from a file and identifies vowel occurrences. Your program should:

- **Read** text from a file named "input.txt".
- **Count** how many times each **vowel (a, e, i, o, u)** appears in the text (case-insensitive).
- **Identify** and display the **most frequently occurring vowel** in the text.
- **Save** the vowel counts and the most frequent vowel in a file named "vowel_count.txt".
- **Display** the results on the console as well.

Example:

Input File (input.txt):

Programming is fun and interesting.

I love learning new things.

Expected Output File (vowel_count.txt):

Vowel Frequency Analysis

a: 3

e: 5

i: 6

o: 3

u: 2

Most frequent vowel: i (6 times)

Expected Console Output:

Vowel count saved in vowel_count.txt

Most frequent vowel: i (6 times)

Additional Instructions:

- The program should be **case-insensitive** (i.e., 'A' and 'a' should be counted together).
- The output should be formatted properly for readability.
- If two vowels have the same highest frequency, display BOTH of them.
- If the input file is empty, display a message "No vowels found in input file."